

USER MANUAL

ESP32 MiniPLC

Part No. 0000001

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Power connection setup

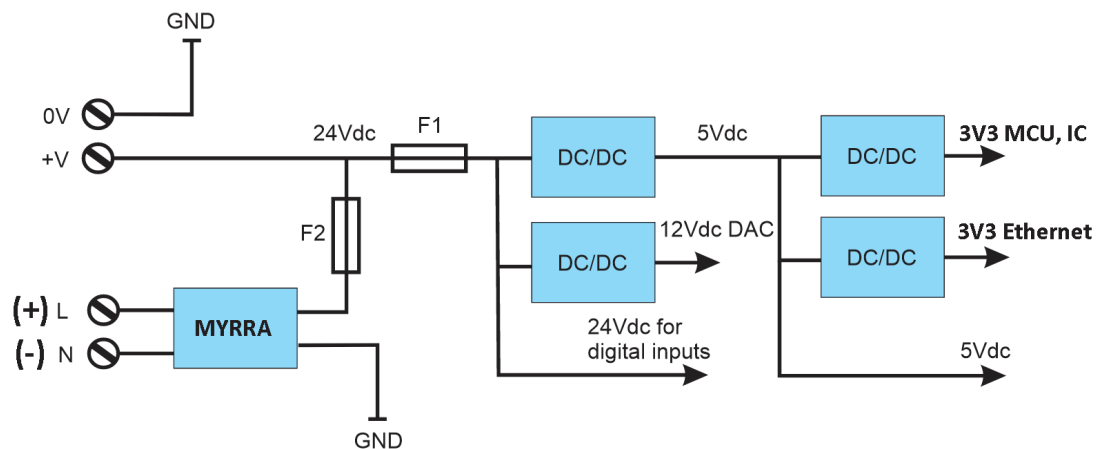
The MiniPLC can be powered either by an external 24VDC power supply or through its internal power supply. It is equipped with an integrated MYRRA 47156 power supply, which accepts 24 V DC at +V/0V, or 85–265 V AC, 47–63 Hz at L/N, operating at a frequency of 47 to 63Hz.

MiniPLC power consumption: typ. 1.85W, max. 3W.

MYRRA 47156 specifications:

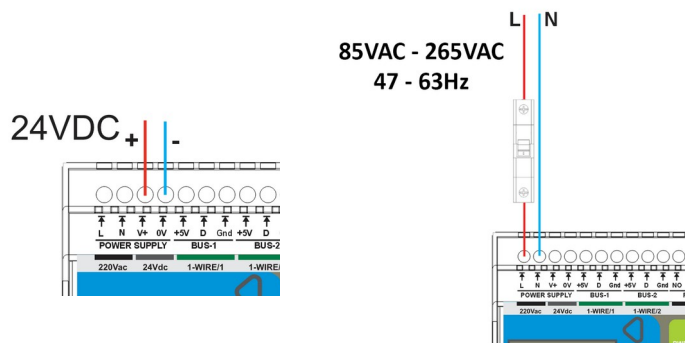
Output voltage (DC Volts)	Output current (DC mA)	Output Power (W)	Efficiency (%)	Ta (°C)
24	220	5	79	+50

The following is a simplified schematic of the internal power distribution:



24 V DC external supply

85–265 V AC via internal power supply

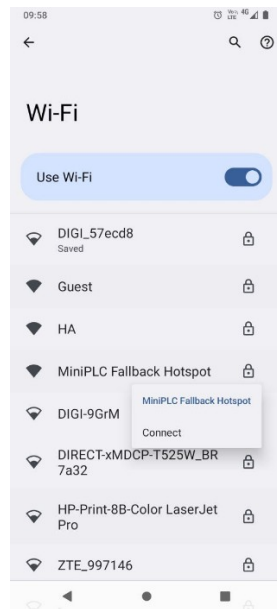


***Do not operate the internal power supply while a 24Vdc external power source is connected to the MiniPLC terminals V+ and 0V.**

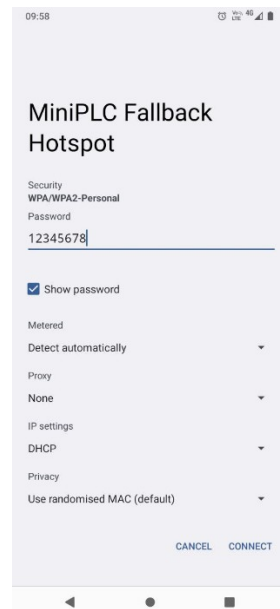
Connecting to the wireless network

To link the MiniPLC to your home Wi-Fi network, start by powering up the MiniPLC, and then use your mobile device or laptop:

1. Connect to "MiniPLC Fallback Hotspot"

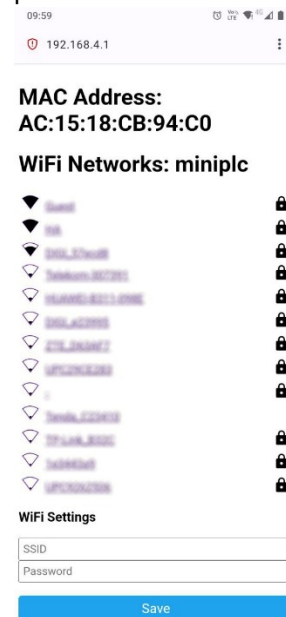


2. Log in using the password "12345678"

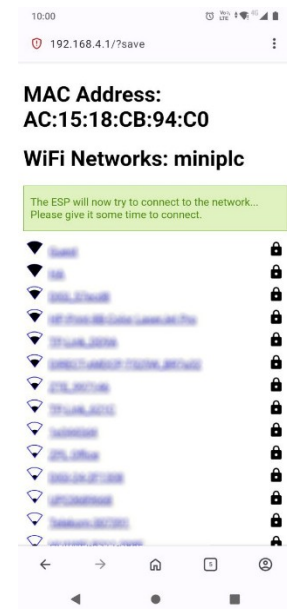


3. Access the URL <http://192.168.4.1>

Choose your home Wi-Fi network and input the password



4. After pressing Save, you can view the assigned address of the MiniPLC in your router's interface.



Web Server

With the factory firmware settings, the MiniPLC includes a built-in web server. You can access it through your web browser by entering the address that your Wi-Fi router has assigned to the MiniPLC.

Build in webserver:

MiniPLC

started 29 minutes 35 seconds ago



Sensor and Control

1-WIRE Dallas temperature BUS1

29.1 °C

1-WIRE Dallas temperature BUS2

29.5 °C

ADC AI1

8.410 V

ADC AI2

0.056 V

ADC AI3

0.046 V

ADC AI4

0.046 V

Button #1

☐

Button #2

☐

Button #3

☐

Button #4

☐

Current time

2025-04-07 15:09

DAC 0-10V

100

0

100

DI #1

☐

DI #2

☐

DI #3

☐

DI #4

☐

LED #2

☐

LED #3

☐

MAX 31856 Temperature 1

22.41 °C

MAX 31856 Temperature 2

23.30 °C

Modbus Sensor

11 C

RELAY #1

☐

RELAY #2

☐

RELAY #3

☐

RELAY #4

☐

RELAY #5

☐

RELAY #6

☐

Switch buzzer

☐

OTA Update

Choose File

NO FILE CHOSEN

UPDATE

Debug Log

Time	Level	Tag	Message
18:09:18	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:18	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:18	[D]	[ads1115.sensor:017]	'ADC AI2': Got Voltage=0.018175V
18:09:18	[D]	[sensor:094]	'ADC AI2': Sending state 0.05512 V with 3 decimals of accuracy
18:09:18	[D]	[ads1115.sensor:017]	'ADC AI3': Got Voltage=0.015000V
18:09:18	[D]	[sensor:094]	'ADC AI3': Sending state 0.04500 V with 3 decimals of accuracy
18:09:19	[D]	[ads1115.sensor:017]	'ADC AI1': Got Voltage=2.004625V
18:09:19	[D]	[sensor:094]	'ADC AI1': Sending state 8.41388 V with 3 decimals of accuracy
18:09:19	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:19	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:19	[D]	[ads1115.sensor:017]	'ADC AI4': Got Voltage=0.015000V
18:09:19	[D]	[sensor:094]	'ADC AI4': Sending state 0.04500 V with 3 decimals of accuracy
18:09:19	[D]	[dallas.temp.sensor:054]	'1-WIRE Dallas temperature BUS2': Got temperature=29.7°C
18:09:19	[D]	[sensor:094]	'1-WIRE Dallas temperature BUS2': Sending state 29.68750 °C with 1 decimals of accuracy
18:09:19	[D]	[dallas.temp.sensor:054]	'1-WIRE Dallas temperature BUS1': Got temperature=29.1°C
18:09:19	[D]	[sensor:094]	'1-WIRE Dallas temperature BUS1': Sending state 29.12500 °C with 1 decimals of accuracy
18:09:20	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:20	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:21	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:21	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:22	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:22	[D]	[text_sensor:064]	'Current time': Sending state '2025-04-07 15:09'
18:09:23	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:23	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:23	[D]	[ads1115.sensor:017]	'ADC AI2': Got Voltage=0.018562V
18:09:23	[D]	[sensor:094]	'ADC AI2': Sending state 0.05569 V with 3 decimals of accuracy
18:09:23	[D]	[ads1115.sensor:017]	'ADC AI3': Got Voltage=0.015188V
18:09:23	[D]	[sensor:094]	'ADC AI3': Sending state 0.04556 V with 3 decimals of accuracy
18:09:24	[D]	[ads1115.sensor:017]	'ADC AI1': Got Voltage=2.003313V
18:09:24	[D]	[sensor:094]	'ADC AI1': Sending state 8.40994 V with 3 decimals of accuracy
18:09:24	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:24	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy
18:09:24	[D]	[ads1115.sensor:017]	'ADC AI4': Got Voltage=0.015188V
18:09:24	[D]	[sensor:094]	'ADC AI4': Sending state 0.04556 V with 3 decimals of accuracy
18:09:24	[D]	[dallas.temp.sensor:054]	'1-WIRE Dallas temperature BUS2': Got temperature=29.3°C
18:09:24	[D]	[sensor:094]	'1-WIRE Dallas temperature BUS2': Sending state 29.50000 °C with 1 decimals of accuracy
18:09:24	[D]	[dallas.temp.sensor:054]	'1-WIRE Dallas temperature BUS1': Got temperature=29.1°C
18:09:24	[D]	[sensor:094]	'1-WIRE Dallas temperature BUS1': Sending state 29.06250 °C with 1 decimals of accuracy
18:09:25	[D]	[modbus_controller.sensor:025]	Sensor new state: 11.00
18:09:25	[D]	[sensor:094]	'Modbus Sensor': Sending state 11.00000 C with 0 decimals of accuracy

Connecting to the Home Assistant platform

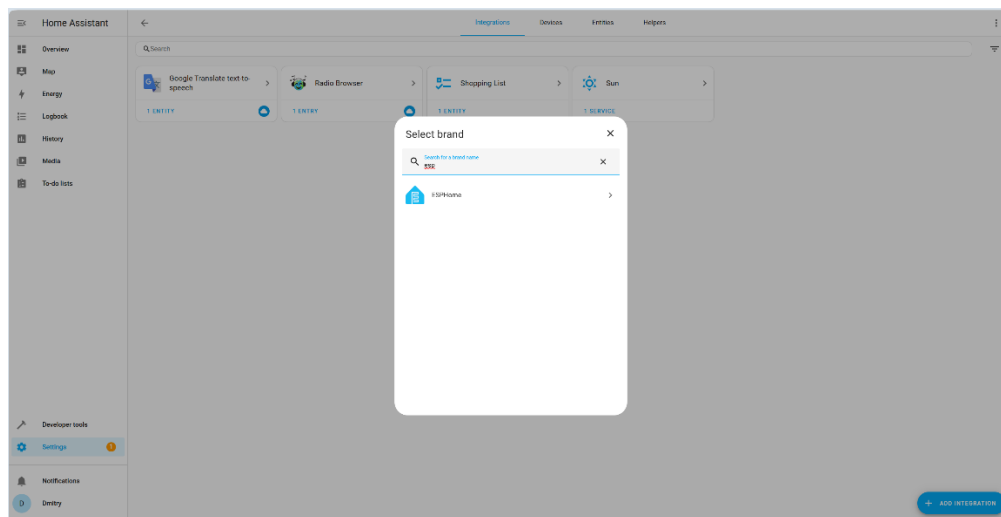
To connect the MiniPLC to your Home Assistant setup, follow these steps:

1.Power the MiniPLC

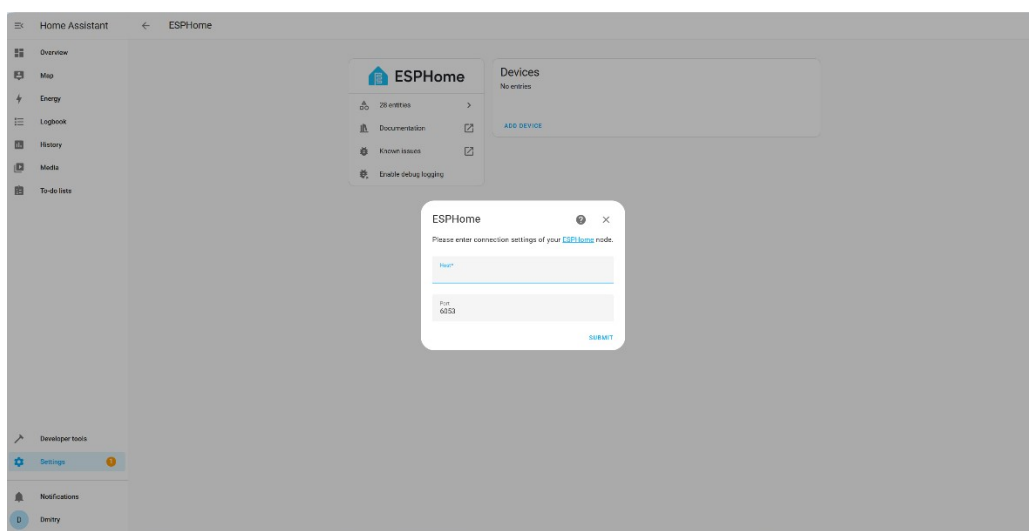
2.Connect to Wi-Fi

3.Integrate with Home Assistant:

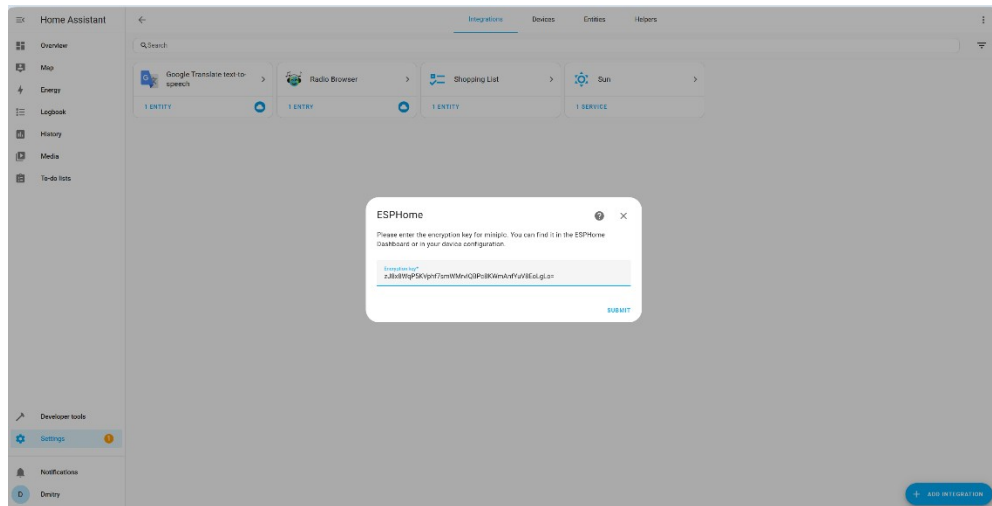
Setup ESPHome integration in Home Assistant:



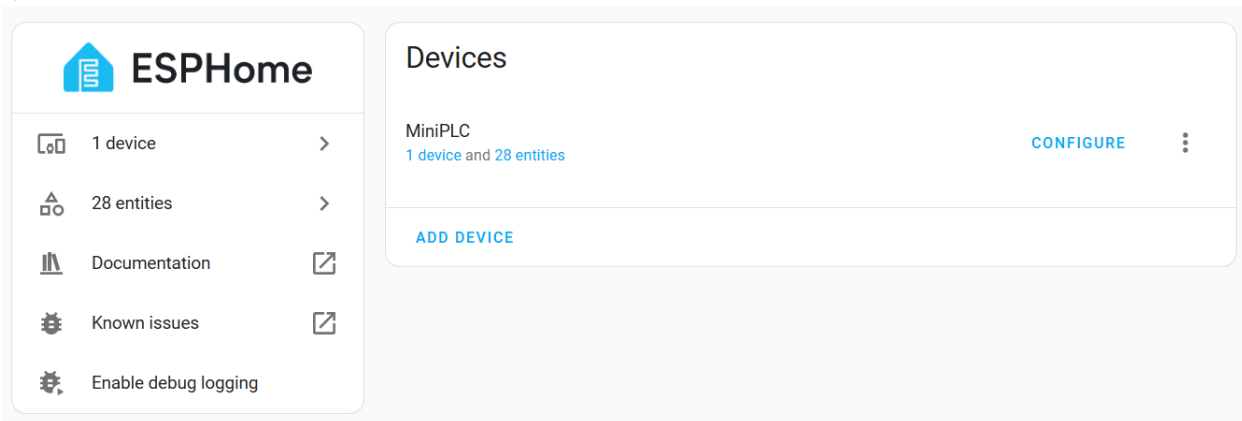
In the ESPHome integration press "Add device" and enter the address that your Wi-Fi router has assigned to the MiniPLC:



Input the encryption key from your configuration file, or the default encryption key provided in the factory setup firmware "zJ8x8WqP5KVphf7smWMrvlQBpc8KWmAnfYuV8EoLgLo=":



By following these steps, you can effectively connect and integrate the ESP32 MiniPLC with your Home Assistant system:



Controls

-  DAC 0-10V ☐
-  LED #2 ☐
-  LED #3 ☐
-  RELAY #1 ☐
-  RELAY #2 ☐
-  RELAY #3 ☐
-  RELAY #4 ☐
-  RELAY #5 ☐
-  RELAY #6 ☐
-  Switch buzzer ☒

[ADD TO DASHBOARD](#)

Sensors

-  1-WIRE Dallas temperature... 29.1 °C
-  1-WIRE Dallas temperature... 29.5 °C
-  ADC AI1 8.412 V
-  ADC AI2 0.055 V
-  ADC AI3 0.044 V
-  ADC AI4 0.047 V
-  Button #1 Off
-  Button #2 Off
-  Button #3 Off
-  Button #4 Off
-  Current time 2025-04-07 15:01
-  DI #1 Off
-  DI #2 Off
-  DI #3 Off
-  DI #4 Off
-  MAX 31856 Temperature 1 22.45 °C
-  MAX 31856 Temperature 2 23.36 °C
-  Modbus Sensor 11 C

[ADD TO DASHBOARD](#)

Firmware Download

Pre-installed Firmware:

The ESP32 MiniPLC comes with ESPHome firmware pre-installed, meaning it's ready for integration with Home Assistant right out of the box.

Reprogramming MiniPLC: You can reprogram or update the settings via ESPHome using the USB Type-C interface, wireless or manually via OTA in the embedded web server.

Downloading firmware via USB:

In the ESPHome dashboard add a New device

Skip the test:

- "MiniPLC"

New device

Name*

MiniPLC

This device will be configured to connect to the Wi-Fi network stored in your secrets.

CANCEL

NEXT

Installation

ESPHome will now create your configuration and install it on your device.

Connect your ESP8266 or ESP32 with a USB cable to your computer and click on connect. You need to do this once. Later updates install wirelessly. [Learn more](#)

Skip this step to install it on your device later or if you are using a Raspberry Pi Pico.

SKIP THIS STEP

CONNECT

Select the device type - ESP32:

Select your device type

Select the type of device that this configuration will be installed on.

ESP32

>

ESP32-S2

>

ESP32-S3

>

ESP32-C3

>

ESP8266

>

Raspberry Pi Pico W

>

BK72xx

>

RTL87xx

>

☒ Use recommended settings

CANCEL

Choose skip installation:



Configuration created!

You can now install the configuration to your device. The first time this requires a cable.

Once the device is installed and connected to your network, you will be able to manage it wirelessly.

Each ESPHome device has a unique encryption key to talk to other devices. You will need this key to include your device in Home Assistant. You can find the key later in the device menu.

Encryption key

/8soHonSOBc4FiFRcjRivZXHlsRyGwYzIK2g=

SKIP

INSTALL

Modify the miniplc.yaml file according to your requirements (you can refer to the default yaml configuration file available on our GitHub page as a guide

<https://github.com/isystemsautomation/HOMEMASTER/tree/main/Firmware/MiniPLC>):

```
miniplc.yaml
274  - platform: gpio
275    reference_resistor: 9000 Ω
276    rtd_resistor_resistor: 1000 Ω
277    update_interval: 60s
278    platform: adxl345
279    multiplatform: "adxl345"
280    gain: 0.144
281    name: "ADXL345"
282    update_interval: 5s
283    filters:
284      - multiply: 1
285    - platform: adxl345
286    multiplatform: "adxl345"
287    gain: 0.144
288    name: "ADXL345"
289    update_interval: 5s
290    filters:
291      - multiply: 1
292    - platform: adxl345
293    multiplatform: "ADXL345"
294    gain: 0.144
295    name: "ADXL345"
296    update_interval: 5s
297    filters:
298      - multiply: 1
299    - platform: adxl345
300    multiplatform: "ADXL345"
301    gain: 0.144
302    name: "ADXL345"
303    update_interval: 5s
304    filters:
305      - multiply: 1
306    - platform: modbus_controller
307    name: "Modbus Controller"
308    modbus_controller: "modbus_controller"
309    register_type: holding
310    address: 0x0001 # Address of the register inside the Modbus slave device
311    unit_of_measurement: "°C"
312    value_type: U_M000
313
314  outputs:
315    - platform: gpio
316      pin: GPIO2
317      id: buzzer_output
318      platform: esp8266
319      id: buzzer_output
320      status_led:
321        pin:
322          - pin: GPIO4
323            number: 7
324            mode:
325              output: true
326              inverted: false
```

Install the settings through a USB cable:

How do you want to install miniplc.yaml on your device?

Wirelessly
Requires the device to be online

Plug into this computer
For devices connected via USB to this computer

Plug into the computer running ESPHome Device Builder
For devices connected via USB to the server

Manual download
Install it yourself using ESPHome Web or other tools

CANCEL

http://127.0.0.1:6052 wants to connect to a serial port

CP2102N USB to UART Bridge Controller (COM5) - Paired



Connect

Cancel

4%

Installing

This will take 2 minutes.
Keep this page visible to prevent slow down

Wiring diagrams:

1-WIRE

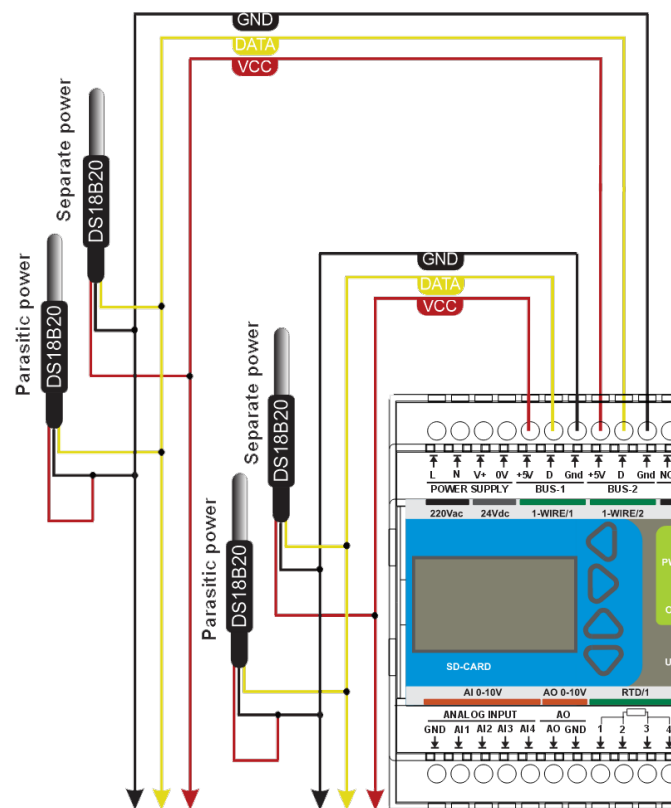
It is advisable to connect a maximum of 20 sensors to the 1-wire bus for optimal performance.

While it is technically feasible to connect up to 32 sensors, doing so will necessitate more complex wiring, which may result in reduced cable length and longer polling intervals.

For the signal lines DATA and GND, utilize a twisted pair from a shielded CAT5/6/7 cable. The VDD line can be connected to an additional conductor within the same cable.

Please ensure that the cable shield remains unconnected, and refrain from transmitting any other data signals through the same cable.

Connection Diagram:



The maximum permissible length of a 1-wire bus is contingent upon the chosen wiring topology. In a linear configuration and a bus topology with short taps, the length can extend up to 300 meters. Conversely, for bus topologies featuring long taps and star configurations, the

limit is reduced to 100 meters. When sensors are connected in a parasitic manner, the overall length must also not exceed 100 meters.

Numerous 1-wire sensors are designed to operate with parasitic power supply, allowing them to draw power through the data line (DATA) while the sensor's VDD is linked to GND. This configuration enables the elimination of one conductor, albeit at the cost of the maximum allowable length of the 1-wire bus.

Relay Output

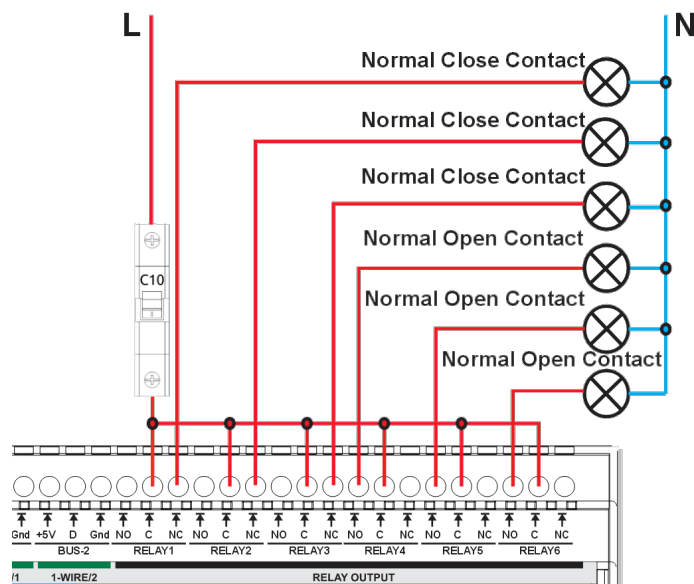
MiniPLC equipped with 6 digital outputs (incorporating relays with optocoupler and varistor for enhanced surge protection)

relays: Hongfa HF115F/005-1ZS3

dry contact specifications:

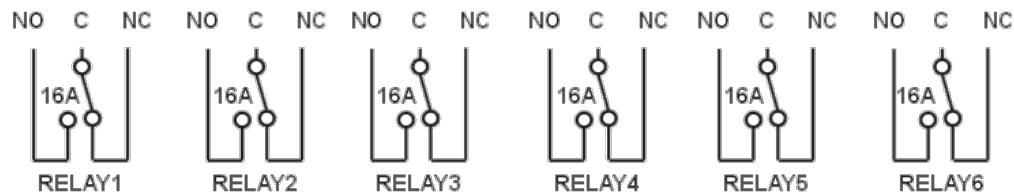
- 3 A @ 250 VAC per contact (module limit)
-
- 30VDC 10A

Connection example:



Depending on the power and type of load, it is necessary to select a protective circuit breaker. At high power of consumers, it is necessary to install a separate protective circuit breaker on each relay. Or use a contactor.

Relay contacts inside MiniPLC:



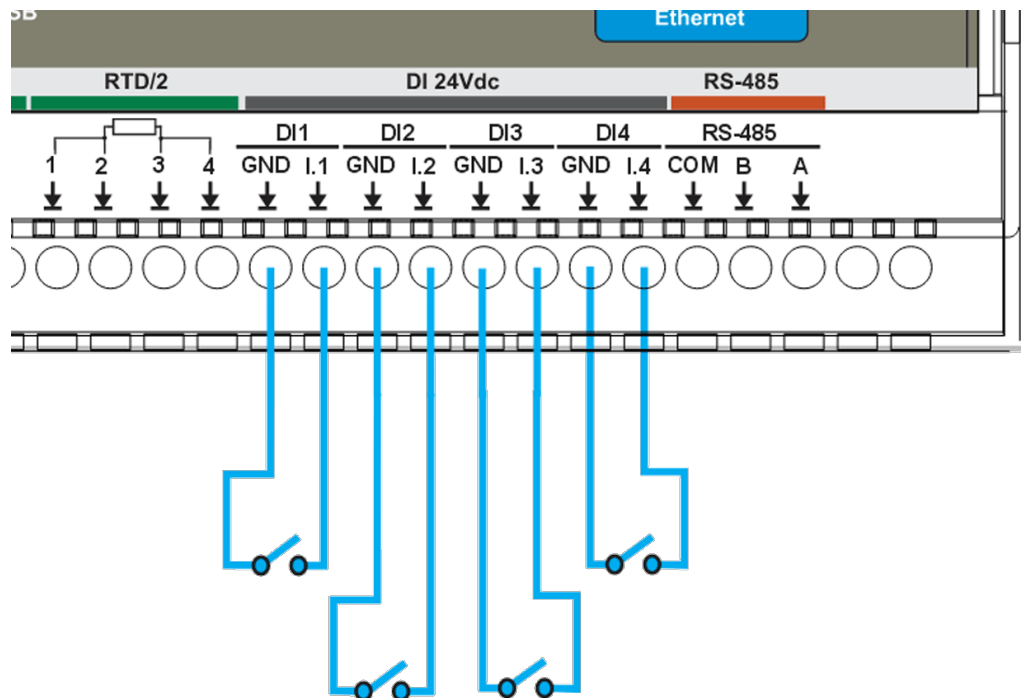
C* Each relay contact is protected from overvoltage by a varistor connected to contact "C"

Digital Inputs

MiniPLC equipped with 4 isolated, surge protected digital sourcing inputs:

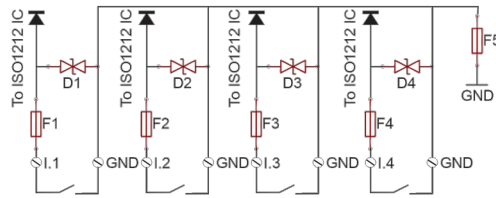
- logic 0: 0 ... 9.2VDC
- undefined: 9.2 ... 15.8VDC
- logic 1: 15.8 ... 24VDC

Connection example:



Digital inputs are designated as source inputs. It is important to note that digital input channels are not intended to serve as a power supply for any devices

Fuses and TVS diodes within the MicroPLC:

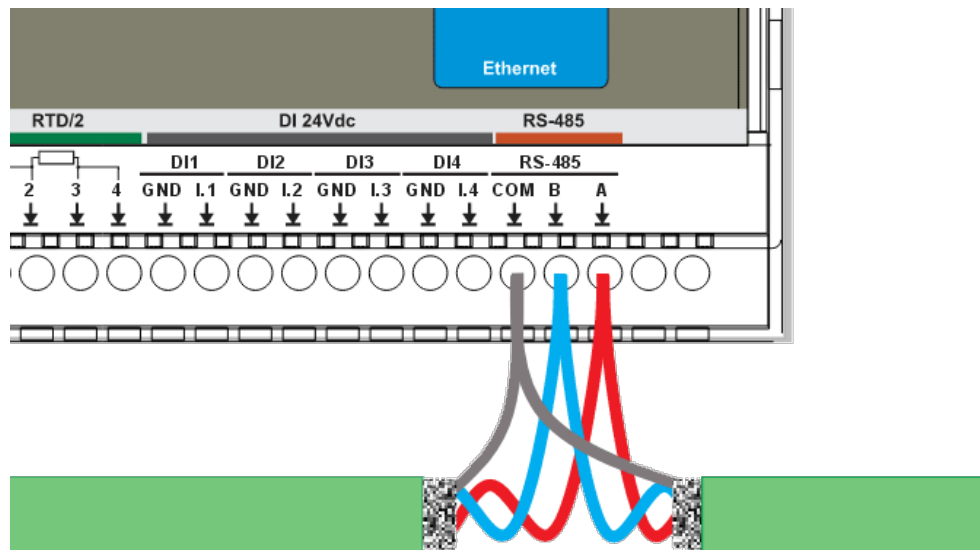


RS-485

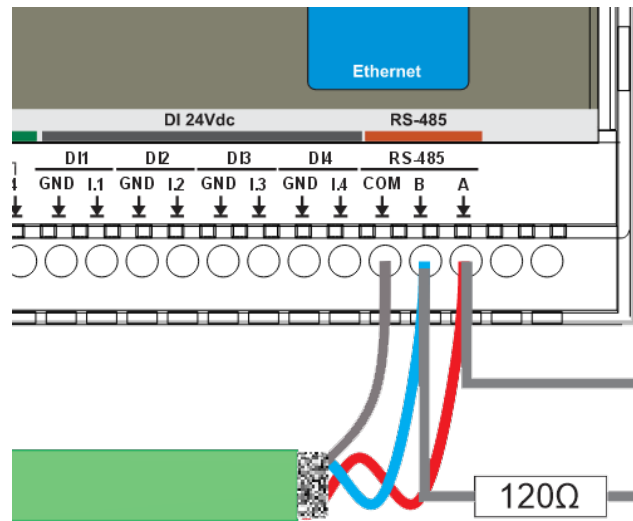
The RS-485 port is not galvanically isolated (MAX485; transceiver shares logic ground). Run COM to every node.

In RS-485 communication systems, it is essential to utilize a shielded twisted pair cable to ensure optimal performance. A notable example of such a cable is the Belden 3105A, compliant with standards EN:50290-2-23 and IEC:60502. However, any cable exhibiting comparable specifications can effectively facilitate the interconnection of all devices within the network.

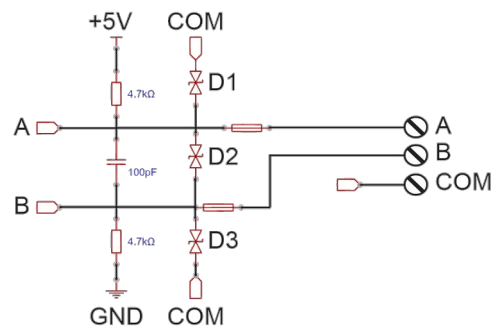
Connection example:



To avoid signal reflections, it is necessary to connect 120 Ohm termination resistors to the ends of the communication cable.:



Automatic fuses and TVS diodes inside MiniPLC for protection:



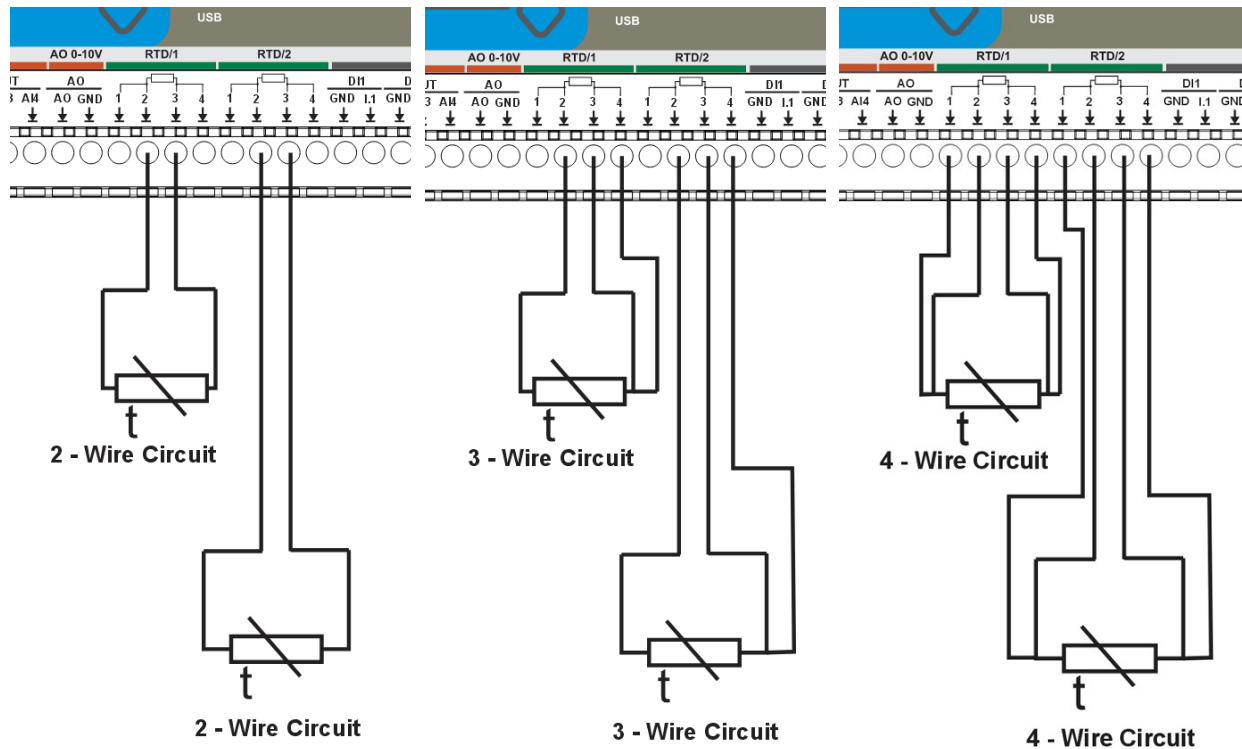
RTD inputs

MiniPLC equipped with the MAX31865, capable of measuring 100Ω to 1kΩ Platinum RTDs (PT100 to PT1000) at 0°C. It supports 2-, 3-, and 4-wire sensor connections, features a 15-bit ADC resolution, and offers a nominal temperature resolution of 0.03125°C (which may vary due to RTD nonlinearity). The total accuracy under all operating conditions is a maximum of 0.5°C

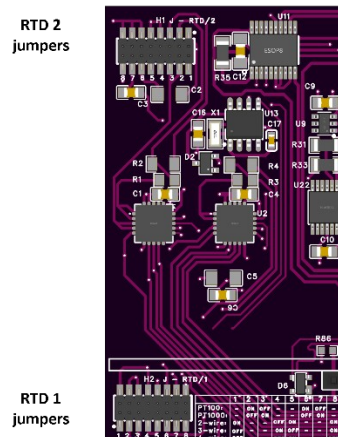
(0.05% of full scale). It includes fully differential VREF inputs, a maximum conversion time of 21ms, and $\pm 45V$ input protection.

Additionally, it provides fault detection for open RTD elements, shorted RTDs, out-of-range voltage, or shorts across the RTD element

Connection example:



To configure the RTD for 2, 3, or 4 wire circuits with either PT100 or PT1000 types, it is essential to adjust the jumper settings on the MCU board:



The factory's default jumper setup is configured for a 2-wire PT100 RTD type for the RTD/1 channel, and a 2-wire PT1000 RTD type for the RTD/2 channel:

	J1	J2	J3	J4	J5	J6	J7	J8
PT100	-	ON	OFF	-	-	ON	OFF	-
PT1000	-	OFF	ON	-	-	OFF	ON	-
2-WIRE	ON	-	-	OFF	ON	-	-	ON
3-WIRE	OFF	-	-	ON	OFF	-	-	ON
4-WIRE	OFF	-	-	OFF	ON	-	-	OFF

Additionally, it is essential to configure the reference resistor in the YAML file:

```
#MAX 31865 RTD
```

```
- platform: max31865
```

```
  name: "MAX 31856 Temperature 1"
```

```
  cs_pin: GPIO1
```

```
#Reference resistance for PT100
```

```
  reference_resistance: 400  $\Omega$ 
```

```
  rtd_nominal_resistance: 100  $\Omega$ 
```

```
  update_interval: 60s
```

```
- platform: max31865
```

```
  name: "MAX 31856 Temperature 2"
```

```
  cs_pin: GPIO3
```

```
#Reference resistance for PT1000
```

```
  reference_resistance: 4000  $\Omega$ 
```

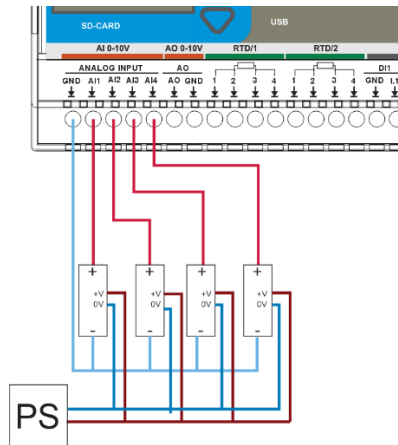
```
  rtd_nominal_resistance: 1000  $\Omega$ 
```

```
  update_interval: 60s
```

0-10V Analog inputs

MiniPLC integrated with the ADS1115, providing 4 analog inputs from 0 to 10VDC, featuring a 16-bit resolution through an operational amplifier, a maximum sampling rate of 0.86 ksp/s, ESD ratings of 2kV HBM and 1.5kV CDM, and includes EMI and RF filtering.

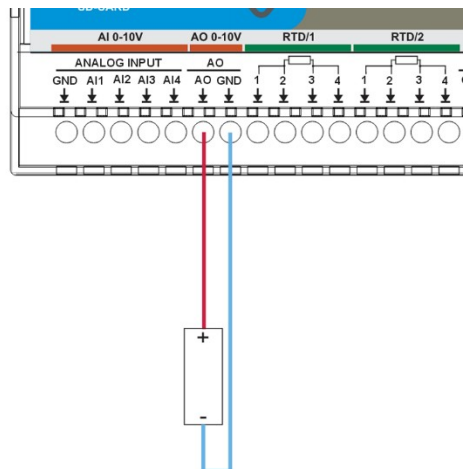
Connection example:



0-10V Analog output

A MiniPLC combined with the MCP4725, delivering an analog output range of 0 to 10VDC, with a resolution of 12 bits achieved through an operational amplifier.

Connection example:



Bus system configuration

I2C:

- sda: GPIO32
- scl: GPIO33

I2C addresses:

- pcf8574/2: 0x38
- pcf8574/1: 0x39
- ads1115: 0x48
- SH1106 128x64: 0x3C
- pcf8563: 0x51

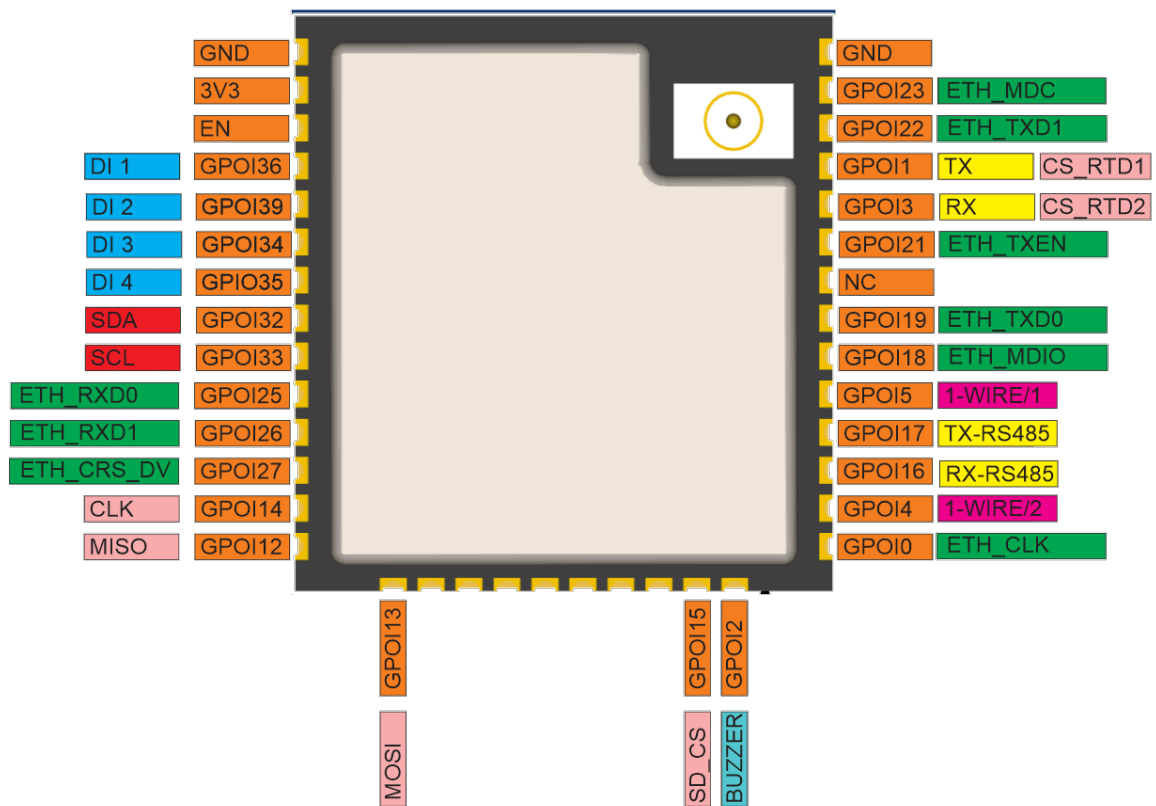
SPI:

- miso_pin: GPIO12
- mosi_pin: GPIO13
- clk_pin: GPIO14

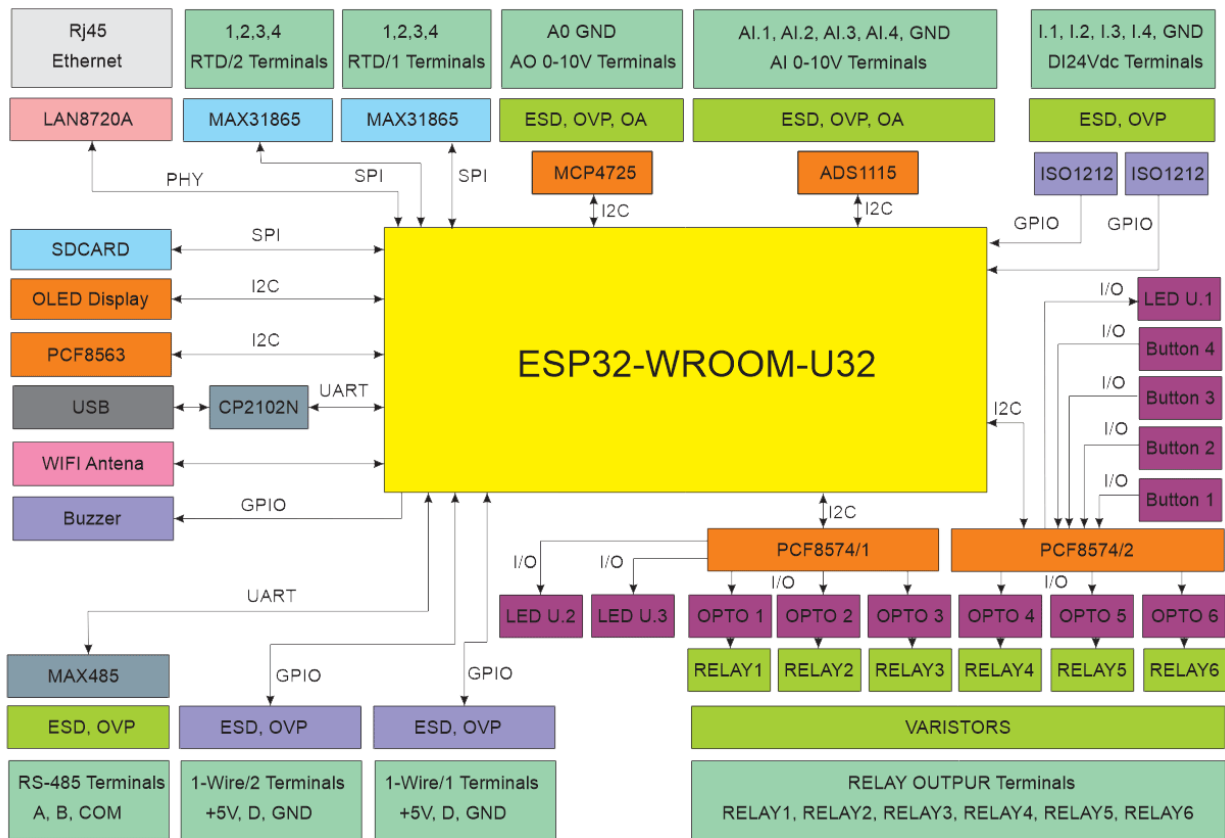
SPI CS pins:

- max31865 RTD1: GPIO01
- max31865 RTD2: GPIO03
- SDCARD: GPIO15

ESP32 Pinout



MiniPLC Function Block Diagram



MiniPLC boards

